

DTSA 5733

Relational Database Design

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	AI Engineer (29.0 / 100)	19.9 / 100
SPECIALIZATION Databases for Data Scientists		

20

CURRENT AVG SCORE
across 6 job roles

73

POTENTIAL SCORE
with recommended additions

+60

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course will prepare students with the tools needed to design a Relational Database System. Formerly offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	29.0	76.2	+47.2	HIGH
Machine Learning Engineer	21.8	81.9	+60.1	HIGH
ML Engineer	17.9	76.3	+58.4	HIGH
MLOps Engineer	17.6	77.2	+59.6	HIGH
Data Scientist	17.5	75.0	+57.5	HIGH
Applied Scientist	15.5	54.3	+38.8	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

Model Deployment & Serving 2 of 6 roles

production

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.
- MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.
- LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

AI Engineer

29.0 → 76.2 (+47.2 pts)

perception

data

engineering

models

learning

infrastructure

build

form

Machine Learning Engineer

21.8 → 81.9 (+60.1 pts)

data

engineering

learning

solutions

exploratory

business

time

development

ML Engineer

17.9 → 76.3 (+58.4 pts)

data

learning

systems

models

production

engineering

starting

infrastructure

MLOps Engineer

17.6 → 77.2 (+59.6 pts)

infrastructure

data

production

systems

learning

engineering

training

models

Data Scientist

17.5 → 75.0 (+57.5 pts)

data

product

characteristics

presentation

evolution

insights

different

social

Applied Scientist

15.5 → 54.3 (+38.8 pts)

lab

models

data

learning

systems

engineering

development

science

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.